

THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENTS
JOINT MID-TERM EXAMINATIONS BIOLOGY FORM FOUR
MARCH-2026

Time: 2:30 Hours



INSTRUCTIONS

1. This paper consists of sections A, B and C with a total of eleven (11) questions.
2. Answer all questions in sections A and B and two (2) questions from section C.
3. Section A carries sixteen (16) marks, section B fifty four (54) marks and section C thirty (30) marks.
4. Except for drawings which must be drawn in pencil, all writing must be in blue or black ink.
5. Cellular phones and any unauthorized materials are not allowed in the examination room.
6. Write your name on every page of your answer booklet(s).

SECTION A (16 Marks)

1. For each of the items (i–x) choose the correct answer from the alternatives given.
 - (i) In a garden of peas, a gardener observed the seedlings with its cotyledons on the ground. Those seedlings undergo;
A. Hypogeal germination B. Epigeal germination
C. Cotyledon germination D. Positive growth
E. Dicotyledon.
 - (ii) One of the following is the similarity between Division Coniferophyta and Division Angiospermophyta;
A. They reproduce sexually by seeds B. They are wind pollinated
C. They have cones D. They produce leaves
E. They produce flowers.

(iii) Mr. Chipotela started salivating as he saw peeled half ripe mango with topping of red chilli powder. What part of the brain is concerned with this salivation?

A. Hypothalamus B. Medulla oblongata

C. Cerebellum D. Cerebrum

E. Cerebral hemisphere.

(iv) When giving First Aid to a snake bite victim you should not;

A. Cut the wound B. Calm them down

C. Remove jewellery from the bitten limb

D. Lower the bitten area E. Wash the wound.

(v) Viruses are considered to be non-living because;

A. They have true nuclei B. They are only active in the contents of a living cell

C. The body is covered by cell wall D. They are single cell eukaryotic organism

E. They have cell wall made up of cellulose.

(vi) A blood group B patient hospitalized for blood deficiency can only receive blood from two possible donors;

A. B+ or O+ B. A- or O+

C. B- or O- D. B+ or B-

E. B- or AB.

(vii) When students were conducting private study at night, light went off suddenly. Which change occurred in the eye?

A. The lens becomes wider B. The pupil become large

C. The ciliary muscle relaxed D. The lids closed

E. The eye opened wider.

(viii) One of the following is the principle of good manners;

A. Eating fast B. Speaking respectfully

C. Throwing food D. Wearing light clothes

E. Washing the body every day.

(ix) The following is a set of growth regulating hormones in plants;

- A. Auxin, cytokinin, gibberellin and abscisic acid
- B. Auxin, ethylene, gibberellin and ptyalin
- C. Cytokinin, ethylene, abscisic acid and folinic acid
- D. Gibberellin, auxin, cytokinin and formalin.

(x) Crossing unknown genotype of an individual with a homozygous recessive parent is regarded as;

- A. Monohybrid cross B. Dihybrid cross
- C. Back cross D. Test cross E. Vigor cross.

Question 2: Matching items

LIST A	LIST B
(i) An opening that allows entry of water and oxygen into a seed during germination. (ii) Inability of a seed to germinate. (iii) Prevent a viable seed to germinate. (iv) A type of cell division whereby the same number of chromosomes is maintained from parent cell to daughter cell. (v) A type of seed germination whereby cotyledons are pushed above the ground. (vi) Located at the tips of roots and shoots.	A. Micropyle B. Hilum C. Meristematic zone D. Elongation zone E. Viability F. Dormancy G. Epigeal germination H. Hypogeal germination I. Meiosis J. Mitosis K. Hard seed coat

SECTION B (54 Marks)

Answer all questions in this section.

3. (a) Makoti is a civil engineering student at a certain university doing research for construction of a good school laboratory. Explain four qualities to consider when constructing a good laboratory.

(b) Joseph was bitten by a snake and given immediate care before going to hospital.

(i) What is the name of the immediate care?

(ii) Mention four things that should be done.

4. Juma has prolonged coughing, blood stained sputum, poor appetite, weight loss and night sweating.

(a) Which disease is he likely suffering from?

- (b) What causes the disease?
- (c) How can it be prevented? (Four points)
- (d) What advice would you give Juma?

5. A headmaster lifted a glass of very hot coffee and suddenly dropped it due to extreme heat.

- (i) Name the action that occurred in his body.
- (ii) Name the pathway followed.
- (iii) Draw a labelled diagram showing components involved.
- (iv) State the significance of this action.

6. (a) Earthworms are called farmer's friends. Give three reasons.

(b) Why are earthworms said to be hermaphrodites?

(c) Give four features that make arthropods successful organisms on earth.

7. (a) Give reasons:

- (i) Respiratory surfaces must be moist.
- (ii) The ileum is long and folded.
- (iii) Laboratory floors are rough.
- (b) State three functions of the skin.

8. During interclass competition a player suddenly felt severe pain in his leg muscles.

(a) Suggest the problem.

(b) State two causes.

(c) Give two preventive measures.

SECTION C (30 Marks)

Answer any TWO questions.

9. As a guest in a workshop on Birth Control Methods, explain methods which:

- (a) Prevent union of gametes (4 points)
- (b) Suppress formation or release of gametes (2 points)
- (c) Prevent implantation of the zygote.

10. Discuss factors contributing to risky behaviours among young people and their consequences.

11. Explain six factors that can make a person look older than his or her real age.